

Notes: Kaniel, Saar, and Titman (JF, 2008)

Individual Investor Trading and Stock Returns

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1 Big picture

- **Question.** How is net trading by individual investors related to short-horizon stock returns? Do individuals simply chase returns, or does their aggregate trading help explain next-week and next-month return predictability?
- **Setting.** The paper uses a proprietary NYSE data set from 2000–2003 covering all common domestic stocks traded on the NYSE, matched to CRSP. The key advantage is that the data identify aggregate individual-investor buy and sell volume directly from executed orders.
- **Main object.** The central variable is *Net Individual Trading* (NIT): individual buy dollar volume minus individual sell dollar volume, scaled by the stock’s average dollar volume over the prior year.
- **Core empirical pattern.** Individuals are short-horizon contrarians. They tend to buy stocks after recent price declines and sell stocks after recent price increases.
- **Main return result.** Stocks with intense individual buying earn positive abnormal returns over the next month, while stocks with intense individual selling earn negative abnormal returns. In the authors’ headline event-window results:
 - the 20 days before intense individual selling have a market-adjusted return of about +3.15%;
 - the 20 days before intense individual buying have a market-adjusted return of about –2.47%;
 - the 20 days after intense individual buying have a market-adjusted return of about +0.80%;
 - the 20 days after intense individual selling have a market-adjusted return of about –0.33%.
- **Main interpretation.** The authors argue that individuals are not necessarily better informed. Instead, their contrarian behavior makes them implicit liquidity providers to institutions that demand immediacy. Institutions push prices down when they urgently sell and up when they urgently buy; individuals take the other side and earn compensation through subsequent partial reversals.
- **Why this is not just old return reversal.** The paper shows that NIT predicts future returns even after controlling for past returns and turnover. In double sorts, NIT remains predictive while past weekly returns do not have an independent reversal effect. In Fama–MacBeth regressions, NIT remains strongly significant.
- **Why liquidity provision is plausible.** The return effect is larger in less liquid stocks. The contemporaneous return during the intense-trading week has the opposite sign from the subsequent reversal: prices fall when individuals intensely buy and rise when individuals intensely sell. This is exactly the sign pattern of price pressure plus reversal.
- **What the paper is not claiming.** The paper does not claim that the average individual investor necessarily earns large abnormal long-run returns. The documented predictability is short-horizon and portfolio-level. Individual holding periods may be much longer than the one-month reversal window.
- **Economic takeaway.** Individual investors can look like “noise traders” in many contexts, but in this setting their contrarian trading helps absorb institutional order flow. Their trading is therefore informative about short-horizon liquidity pressure.

2 Data and measurement (key objects)

2.1 NYSE individual-investor order data

- **Source.** The paper uses NYSE Consolidated Equity Audit Trail Data (CAUD). These files contain detailed information on all orders that execute on the exchange, including both electronic orders and manually handled orders.
- **Individual-investor flag.** Each order contains an *Account Type* field. The NYSE created individual-investor categories after the October 1987 market break to support priority delivery of individual-investor orders during extreme conditions.
- **Key advantage.** The buy/sell classification is based on orders, not on an ex post trade-signing algorithm. This matters because the paper does not need to infer whether a trade was buyer-initiated or seller-initiated using the Lee–Ready algorithm.
- **Coverage.** The sample includes 2,034 common domestic NYSE stocks traded at any point from January 1, 2000 to December 31, 2003.
- **Scale.** The data contain roughly \$1.55 trillion of individual-investor trading over the four-year sample.
- **Matched data.** CRSP is used for returns, market capitalization, turnover, and dollar volume. TAQ is used in robustness tests to compute end-of-day quote-midpoint returns and effective spreads.

Interpretation. The data are unusually good for studying individual investors because they observe aggregate retail buying and selling directly across a large cross-section of stocks. Most other work either uses a single broker’s clients or proxies retail trading by small trades.

2.2 Important limitation: NYSE executions only

Some retail orders in NYSE-listed stocks are internalized by brokers or routed to regional exchanges. Those executions do not appear in the NYSE execution data.

The authors argue that this is not fatal for their purpose:

- balanced retail order flow is easier for brokers to internalize;
- imbalanced retail order flow is more likely to reach the NYSE;
- the paper’s key variable is an imbalance measure, not total retail participation.

Interpretation. The NIT measure may not capture all individual trading, but it is likely to capture meaningful net imbalances in individual trading.

2.3 Size groups

Stocks are sorted by market capitalization into 10 deciles and grouped as follows:

- small stocks: size deciles 1–4,
- mid-cap stocks: size deciles 5–7,

- large stocks: size deciles 8–10.

This grouping is used repeatedly because liquidity, volatility, individual order size, and return predictability differ strongly by size.

2.4 Net Individual Trading measure

For stock i on day t , the paper defines

$$\text{NIT}_{i,t} = \frac{\text{Individual buy dollar volume}_{i,t} - \text{Individual sell dollar volume}_{i,t}}{\text{Average daily dollar volume over the prior year}_{i,t}}. \quad (1)$$

Interpretation.

- A positive value means individuals are net buyers.
- A negative value means individuals are net sellers.
- Scaling by past dollar volume makes the imbalance comparable across small and large stocks.
- The paper usually aggregates daily NIT to the weekly frequency to match the short-horizon return-predictability literature.

2.5 Why use a stock-specific intensity measure?

A raw value of NIT may be large for one stock but normal for another. The paper therefore defines intense trading relative to each stock’s recent NIT history.

For each week, each stock is placed into a decile by comparing its current weekly NIT with its own NIT values in the previous nine weeks:

- decile 1: current NIT is unusually negative relative to the stock’s own recent history; this is *intense individual selling*;
- decile 10: current NIT is unusually positive relative to the stock’s own recent history; this is *intense individual buying*.

Interpretation. This is not a change-in-NIT measure. The variable is still the level of the imbalance. The previous nine weeks provide the benchmark for deciding whether the level is unusually intense.

2.6 Market-adjusted returns

For most event-window exercises, the paper computes market-adjusted returns by subtracting the return on a market proxy, usually the equal-weighted portfolio of all stocks in the sample.

For a cumulative return window, a useful notation is

$$\text{MAR}_{i,[a,b]} = R_{i,[a,b]} - R_{[a,b]}^{mkt}. \quad (2)$$

The paper then averages these market-adjusted returns across stocks in each intense-trading portfolio and across weeks.

Interpretation. The question is not whether the market rose after retail buying. The question is whether stocks bought or sold intensely by individuals outperform or underperform the relevant market benchmark.

3 Empirical specifications (all equations + interpretation)

3.1 Event-style return dynamics around intense individual trading

Let t be the formation week in which a stock enters an intense NIT portfolio. The paper examines cumulative market-adjusted returns before, during, and after that week:

$$\overline{MAR}_{d,k} = \frac{1}{|P_{d,t}|} \sum_{i \in P_{d,t}} MAR_{i,t+k}, \quad (3)$$

where $P_{d,t}$ is the portfolio of stocks in NIT decile d during week t .

The two most important portfolios are:

- $d = 1$: intense individual selling;
- $d = 10$: intense individual buying.

Interpretation. This event design asks: What happened to prices before individuals traded intensely, what happened during the trading week, and what happened afterward?

3.2 Return predictability with double sorts

To test whether NIT just proxies for known return patterns, the paper forms 5×5 portfolios.

For the past-return test:

- one sort is on previous-week stock return;
- the other sort is on NIT intensity;
- the outcome is the next week's market-adjusted return.

For the turnover test:

- one sort is on previous-week turnover intensity;
- the other sort is on NIT intensity;
- the outcome is the next week's market-adjusted return.

A typical NIT spread portfolio is

$$R_{q,t+1}^{NIT5-NIT1} = R_{NIT=5,q,t+1} - R_{NIT=1,q,t+1}, \quad (4)$$

where q is a return quintile or turnover quintile.

Interpretation. If NIT merely captures past-return reversal, the NIT spread should disappear after sorting on past returns. It does not.

3.3 Fama–MacBeth return regressions

The paper estimates weekly cross-sectional regressions of the form

$$R_{i,t+1} = \alpha_t + \beta_t R_{i,t} + \gamma_t \text{NITDecile}_{i,t} + \delta_t \text{TurnoverDecile}_{i,t} + \varepsilon_{i,t+1}. \quad (5)$$

The reported coefficients are time-series averages:

$$\bar{\gamma} = \frac{1}{T} \sum_{t=1}^T \gamma_t, \quad \bar{\delta} = \frac{1}{T} \sum_{t=1}^T \delta_t. \quad (6)$$

Standard errors use Newey–West corrections.

Interpretation.

- β_t captures short-run return reversal.
- γ_t captures whether NIT predicts next-week returns after controlling for past returns and volume.
- δ_t captures the high-volume return premium.
- The key result is that $\bar{\gamma} > 0$ remains strongly significant.

3.4 Midquote-return robustness

A possible problem with CRSP close-to-close returns is bid–ask bounce. To address this, the paper constructs returns using end-of-day TAQ quote midpoints.

Let $m_{i,t}$ be the closing quote midpoint. The midpoint return is approximately

$$R_{i,t}^{mid} = \frac{m_{i,t} - m_{i,t-1}}{m_{i,t-1}}. \quad (7)$$

Interpretation. If past-return reversal is mostly a bid–ask bounce artifact, it should weaken when returns are computed from midquotes. This is what the paper finds. NIT remains predictive.

3.5 Volatility around intense individual trading

To test whether the NIT return premium is simply compensation for higher risk, the paper computes abnormal volatility around the intense-trading week.

For stock i and event-time window centered at k , define

$$AVOL_{i,k} = SD(R_{i,\tau} : \tau \in \text{9-day window centered at } k) - SD_i^{normal}, \quad (8)$$

where SD_i^{normal} is the average 9-day return standard deviation over nonoverlapping windows in the full sample.

Interpretation. If the subsequent return premium is compensation for a permanent increase in risk, volatility should remain high after intense NIT events. Instead, volatility rises before and during the event and then largely reverses.

3.6 Liquidity test using effective spreads

The liquidity-provision explanation predicts that compensation should be larger in less liquid stocks. The paper uses percentage effective spread as the liquidity proxy:

$$ES_{i,t} = \frac{|P_{i,t} - M_{i,t}|}{M_{i,t}}, \quad (9)$$

where $P_{i,t}$ is the transaction price and $M_{i,t}$ is the quote midpoint.

Stocks are sorted into low-, medium-, and high-spread groups. The NIT event exercise is then repeated within each spread group.

Interpretation. A larger NIT return effect among high-spread stocks supports the idea that NIT identifies liquidity provision rather than information revelation.

3.7 Systematic-noise test using principal components

The paper asks whether individual-investor trading is systematic across stocks. Let X_t be the vector of daily NIT values across stocks. A principal components analysis decomposes the covariance matrix of X_t :

$$\Sigma_X v_j = \lambda_j v_j. \quad (10)$$

The percentage of variance explained by the first K components is

$$\frac{\sum_{j=1}^K \lambda_j}{\sum_j \lambda_j}. \quad (11)$$

Because any finite matrix mechanically has some apparent components, the paper compares the real-data principal components with simulated independent data.

Interpretation. The authors find weak evidence of a systematic common component in individual trading. NIT is useful at the stock level, but it does not appear to be a strong market-wide sentiment factor in this sample.

4 Identification and interpretation (what makes the estimates credible)

4.1 What identifies the empirical effect?

The paper's key identifying variation is weekly, stock-level variation in whether individuals are unusually net buyers or net sellers relative to the same stock's recent history.

This structure helps in several ways:

- it avoids comparing naturally high-retail stocks with naturally low-retail stocks;
- it focuses on unusually intense imbalances, where return effects should be easier to detect;
- it makes the NIT event portfolios robust to slow trends in retail trading intensity;
- it allows the authors to compare return predictability from NIT with predictability from past returns and turnover.

4.2 Why the data are unusually credible

The paper's data advantage is that buy and sell volume are directly classified by the order's account type.

This is important because much of the trading literature must classify trades indirectly. For example, a trade-signing algorithm may identify who used a market order, but that is not the same as identifying whether the ultimate trader is an individual or an institution. Kaniel, Saar, and Titman directly observe individual-investor order flow executed on the NYSE.

4.3 Why the paper favors the liquidity-provision interpretation

The liquidity-provision interpretation fits four facts simultaneously:

- **Contrarian retail trading.** Individuals buy after prior price declines and sell after prior price increases.
- **Price pressure during the event week.** During intense individual buying weeks, prices are negative; during intense individual selling weeks, prices are positive. This suggests individuals take the opposite side of price pressure.
- **Subsequent reversal.** After intense buying, prices rebound; after intense selling, prices fall.
- **Larger effect in illiquid stocks.** If compensation is payment for absorbing order flow, it should be higher where order flow is harder to absorb. The spread-sorted evidence supports this.

4.4 Alternative explanations and why they are less convincing

- **Private information by individuals.** This would imply that individuals buy before good news or sell before bad news. But the event-week price movement goes the wrong way: prices fall when individuals buy intensely and rise when they sell intensely.
- **Investor recognition or attention.** A Merton-style recognition story would suggest that increased individual buying raises prices immediately as the stock becomes more widely held. The paper instead finds negative contemporaneous returns during intense buying weeks.
- **Serial correlation in retail order imbalance.** If continued buying mechanically pushes prices up in the future, then contemporaneous returns should also be positive with buying. Again, the observed contemporaneous sign is opposite.
- **Higher risk.** Volatility rises around the intense-trading event but then largely returns to normal. There is no strong evidence of a permanent increase in risk that could explain the abnormal returns.

4.5 Main limitations

- **NYSE-only execution data.** Some retail trading is internalized or routed away from the NYSE. The authors argue that NYSE executions still capture meaningful imbalances, but the data are not the entire retail market.

- **Short sample.** The main sample is 2000–2003. This is enough for weekly panel evidence but not enough for long-horizon individual performance analysis.
- **No direct institutional order-flow data.** The liquidity interpretation relies on individuals taking the other side of institutional pressure, but the paper does not directly observe the institutions’ motives or full order flow.
- **Portfolio-level predictability is not individual-level profit.** Individuals may not rebalance quickly enough to realize the short-horizon reversal profits documented in the paper.
- **Causal caution.** NIT is not randomly assigned. The evidence is best read as a strong predictive and interpretive result, not as a clean randomized causal effect.

5 Tables and figures

5.1 Table I: Summary statistics

Read: This table describes the sample stocks and the individual-investor trading data.

- The sample contains 2,034 NYSE common domestic stocks.
- In CRSP, average market capitalization is much larger for large stocks than for small stocks, and small stocks have higher weekly return volatility.
- In the NYSE individual-trading data, the average weekly individual dollar volume is about \$4.3 million per stock.
- Individual weekly dollar volume is about \$716,000 in small stocks and over \$11 million in large stocks.
- The mean executed order size for individuals is about \$15,822; the median is about \$13,243.
- Individual order size rises with stock size: order size is much larger in large stocks than in small stocks.

Why it matters. The table establishes that the data are large and broad. This is not a small-broker data set. It also shows why scaling NIT by stock volume is necessary.

5.2 Table II: Summary statistics of individual trading imbalances

Read: This table describes the NIT variable.

Important numbers:

- Average NIT is negative across size quintiles, meaning individuals are net sellers on average during the 2000–2003 sample.
- NIT has substantial time-series dispersion, especially among smaller stocks.
- For intense selling portfolios, average NIT is about -0.4171 for all stocks.
- For intense buying portfolios, average NIT is about $+0.2048$ for all stocks.

Table I
Summary Statistics

The sample of stocks for the study consists of all common domestic stocks that were traded on the NYSE at any time between January 1, 2000, and December 31, 2003, with records in the CRSP database. We use ticker symbol and CUSIP to match the stocks to a special data set containing daily aggregated buying and selling volume of individuals provided by the NYSE. There are 2,034 stocks in our sample. In Panel A we provide summary statistics from the CRSP database. For each stock we compute the following time-series measures: AvgCap is the average monthly market capitalization over the sample period; AvgPrc is the average daily closing price; AvgTurn is the average weekly turnover (number of shares traded divided by the number of shares outstanding); AvgVol is the average weekly dollar volume; and StdRet is the standard deviation of weekly returns. We then sort the stocks by market capitalization into 10 deciles, and form three size groups: small stocks (deciles 1, 2, 3, and 4), mid-cap stocks (deciles 5, 6, and 7), and large stocks (deciles 8, 9, and 10). The cross-sectional mean and median of these measures are presented for the entire sample and separately for the three size groups. In Panel B we compute from the NYSE data set the following time-series measures for each stock: the average weekly Dollar Volume, defined as the sum of executed buy and sell orders, the average weekly Share Volume, and the Executed Order Size of individual investors in terms of both dollars and shares.

Panel A: Summary Statistics of Sample Stocks (from CRSP)					
		AvgCap (in million \$)	AvgPrc (in \$)	AvgTurn (in %)	StdRet (in %)
All stocks	Mean	5,303.2	59.80	2.52	113.18
	Median	943.7	21.98	2.05	21.76
Small stocks	Mean	317.1	13.57	2.37	8.77
	Median	308.7	11.60	1.63	4.55
Mid-cap stocks	Mean	1,311.8	25.94	3.22	42.47
	Median	1,230.3	23.94	2.51	31.23
Large stocks	Mean	14,054.0	136.87	3.10	302.56
	Median	5,018.0	37.15	2.56	159.45

Panel B: Summary Statistics of Individual Trading from the NYSE Data Set				
		Individuals Dollar Volume (1000s \$)	Individuals Share Volume (shares)	Individuals Executed Order Size (dollars)
All stocks	Mean	4,304.1	169,243	15,822.0
	Median	1,131.0	55,575	13,243.4
Small stocks	Mean	716.3	149,812	8,689.6
	Median	377.1	32,475	7,896.8
Mid-cap stocks	Mean	2,144.2	177,000	15,723.4
	Median	1,417.1	55,460	14,323.0
Large stocks	Mean	11,147.6	329,988	26,083.4
	Median	4,991.8	141,125	22,240.8

Table II
Summary Statistics of Individuals' Trading Imbalances

The sample of stocks for the study consists of all common domestic stocks that were traded on the NYSE at any time between January 1, 2000, and December 31, 2003, with records in the CRSP database. We use ticker symbol and CUSIP to match the stocks to a special data set containing daily aggregated buying and selling volume of individuals provided by the NYSE. There are 2,034 stocks in our sample. NIT_{*t*} for a stock on week *t* is defined as the dollar volume bought by individuals minus the dollar volume sold by individuals (both obtained from the NYSE), divided by a moving average of past 1-year average dollar volume (from CRSP). In Panel A we compute the time-series means of NIT for each stock over the 4-year sample period. We then sort stocks into five size quintiles according to average market capitalization (from CRSP), and present cross-sectional summary statistics of the stocks' NIT for the five size quintiles. In Panel B we compute the time-series standard deviation of NIT of each stock over the sample period and present cross-sectional summary statistics for the five size quintiles. Panel C presents summary statistics of NIT in weekly portfolios of stocks that experience intense buying or selling by individuals. For each week in the sample period, we use the previous 9 weeks to form NIT deciles. Each stock is put into 1 of 10 deciles according to the NIT value in the current week relative to its value in the previous 9 weeks. Decile 1 contains the stocks with the most intense selling (negative NIT) while decile 10 contains the stocks with the most intense buying (positive NIT). For each decile, we compute the average NIT for the stocks in the portfolio. We present time-series summary statistics of the average NIT in deciles 1 and 10 for the entire sample and separately for the three size groups.

Panel A: Cross-Sectional Distribution of Stocks' Average NIT							
Size Quintiles	Mean	Std. Dev.	Min	25%	Median	75%	Max
Q1 (small stocks)	-0.0613	0.2208	-2.0811	-0.1245	-0.0318	0.0371	0.6708
Q2	-0.1208	0.2328	-2.6624	-0.1750	-0.0659	-0.0099	0.5938
Q3	-0.1037	0.1904	-2.8527	-0.1355	-0.0581	-0.0165	0.4683
Q4	-0.0715	0.1157	-0.8468	-0.0907	-0.0382	-0.0141	0.2743
Q5 (large stocks)	-0.0505	0.0943	-1.0200	-0.0550	-0.0320	-0.0158	0.0750

Panel B: Cross-Sectional Distribution of Stocks' Standard Deviation of NIT							
Size Quintiles	Mean	Std. Dev.	Min	25%	Median	75%	Max
Q1 (small stocks)	0.4556	0.3489	0.0046	0.2134	0.4010	0.6061	3.2319
Q2	0.3475	0.3417	0.0100	0.1362	0.2636	0.4464	3.9161
Q3	0.2485	0.2723	0.0017	0.0991	0.1739	0.3014	3.7744
Q4	0.1731	0.1941	0.0060	0.0748	0.1197	0.1974	2.2564
Q5 (large stocks)	0.1041	0.1200	0.0119	0.0571	0.0758	0.1124	1.8754

Panel C: Summary Statistics of NIT in Portfolios of Intense Buying or Selling by Individuals						
Portfolio		Mean	Std. Dev.	25%	Median	75%
Intense selling (decile 1)	All stocks	-0.4171	0.1334	-0.4978	-0.4047	-0.3262
	Small stocks	-0.6614	0.2316	-0.8119	-0.6241	-0.4910
	Mid-cap stocks	-0.3487	0.1557	-0.4507	-0.3207	-0.2326
	Large stocks	-0.1537	0.0582	-0.1939	-0.1455	-0.1079
Intense buying (decile 10)	All stocks	0.2048	0.0700	0.1580	0.1970	0.2417
	Small stocks	0.3812	0.1198	0.2911	0.3781	0.4473
	Mid-cap stocks	0.1118	0.0555	0.0784	0.1016	0.1352
	Large stocks	0.0507	0.0279	0.0329	0.0474	0.0657

- The magnitudes are larger among small stocks: intense selling averages about -0.6614 , and intense buying averages about $+0.3812$.

Interpretation. Intense retail imbalances are economically meaningful. A decile-1 or decile-10 event is not a tiny deviation; it is a large net buying or selling episode relative to normal stock volume.

5.3 Table III: Cumulative and weekly market-adjusted returns around intense individual trading

Read: This is the central event-window table.

Key numbers:

- Before intense selling, the prior 20-day market-adjusted return is about $+3.15\%$.
- Before intense buying, the prior 20-day market-adjusted return is about -2.47% .
- During the intense selling week, the market-adjusted return is about $+1.80\%$.
- During the intense buying week, the market-adjusted return is about -1.11% .
- After intense selling, the next 20-day market-adjusted return is about -0.33% .
- After intense buying, the next 20-day market-adjusted return is about $+0.80\%$.

Interpretation. Individuals buy stocks that have recently fallen and sell stocks that have recently risen. But the more important result is the post-event reversal: stocks bought intensely by individuals rise afterward, and stocks sold intensely by individuals fall afterward.

Why it matters. This table motivates the entire paper. It creates the puzzle: why does retail order flow predict subsequent short-horizon returns?

Table III
Returns around Intense Individual Trading

This table presents analysis of market-adjusted returns around intense buying and selling activity of individuals as given by the net individual trading measure (NIT). For each week in the sample period, we use the previous 9 weeks to form NIT deciles. Each stock is put into 1 of 10 deciles according to the NIT value in the current week relative to its value in the previous 9 weeks. Decile 1 contains the stocks with the most intense selling (negative NIT) while decile 10 contains the stocks with the most intense buying (positive NIT). We present the results for four portfolios: (i) decile 1, (ii) decile 1 and 2, (iii) decile 9 and 10, and (iv) decile 10. Let k be the number of days prior to or following the portfolio formation each week. In Panel A, we calculate eight cumulative return numbers for each of the stocks in a portfolio $CR(t-k, t+k)$, where $k \in \{20, 15, 10, 5\}$ days and t is the first day of the formation week, and $CR(t-k, t+k)$, where $k \in \{5, 10, 15, 20\}$ days and t is the last day of the formation week. The return on each portfolio is then adjusted by subtracting the return on a market proxy (the equal-weighted portfolio of all stocks in the sample). We present the time-series mean and t -statistic for each market-adjusted cumulative return measure and for the market-adjusted return during the intense trading week ($k=0$). In Panel B, we present the time-series mean and t -statistic for weekly market-adjusted returns in the 4 weeks around the formation week (i.e., $CR(t-k, t+k)$, where $k \in \{20, 15, 10, 5\}$ days and t is the first day of the formation week, and $CR(t-k, t+k)$, where $k \in \{5, 10, 15, 20\}$ days and t is the last day of the formation week). * indicates significance at the 1% level and ** indicates significance at the 5% level (both against a two-sided alternative). The t -statistic is computed using the Newey-West correction.

Panel A: Cumulative Market-Adjusted Returns around Intense Individual Trading										
Portfolio	Mean	$k=-20$	$k=-15$	$k=-10$	$k=-5$	$k=0$	$k=5$	$k=10$	$k=15$	$k=20$
Intense Selling	Mean	0.0015**	0.0229**	0.0247**	0.0162**	0.0180**	-0.0013**	-0.0020**	-0.0028**	-0.0033**
decile 1	t -statistic	(22.62)	(22.22)	(23.50)	(20.80)	(20.72)	(-2.77)	(-3.01)	(-3.48)	(-3.48)
Selling	Mean	0.0302**	0.0281**	0.0231**	0.0148**	0.0129**	-0.0013**	-0.0019**	-0.0028**	-0.0033**
deciles 1&2	t -statistic	(24.86)	(22.73)	(24.02)	(22.29)	(21.94)	(-2.90)	(-3.48)	(-4.62)	(-4.91)
Buying	Mean	-0.0051**	-0.0234**	-0.0192**	-0.0124**	-0.0087**	0.0025**	0.0042**	0.0055**	0.0066**
deciles 9&10	t -statistic	(-24.12)	(-24.43)	(-24.07)	(-23.85)	(-16.74)	(7.14)	(8.23)	(8.14)	(10.61)
Intense Buying	Mean	-0.0047**	-0.0229**	-0.0200**	-0.0131**	-0.0111**	0.0032**	0.0048**	0.0066**	0.0080**
decile 10	t -statistic	(-22.21)	(-22.61)	(-22.58)	(-22.04)	(-13.90)	(6.97)	(7.22)	(8.09)	(8.71)

Panel B: Weekly Market-Adjusted Returns around Intense Individual Trading										
Portfolio	Mean	$k=-20$	$k=-15$	$k=-10$	$k=-5$	$k=0$	$k=5$	$k=10$	$k=15$	$k=20$
Intense Selling	Mean	0.0018**	0.0051**	0.0084**	0.0162**	0.0180**	-0.0013**	-0.0008	-0.0009*	-0.0005
decile 1	t -statistic	(5.87)	(9.40)	(16.20)	(20.80)	(20.72)	(-2.77)	(-1.64)	(-1.22)	(-1.21)
Selling	Mean	0.0022**	0.0048**	0.0083**	0.0148**	0.0129**	-0.0013**	-0.0007	-0.0009*	-0.0007*
deciles 1&2	t -statistic	(5.86)	(13.01)	(18.91)	(23.29)	(21.94)	(-3.95)	(-1.86)	(-1.32)	(-1.30)
Buying	Mean	-0.0019**	-0.0043**	-0.0070**	-0.0124**	-0.0087**	0.0025**	0.0018**	0.0012**	0.0012**
deciles 9&10	t -statistic	(-6.41)	(-14.04)	(-18.41)	(-23.85)	(-16.74)	(7.14)	(5.80)	(4.07)	(4.70)
Intense Buying	Mean	-0.0012**	-0.0038**	-0.0072**	-0.0121**	-0.0111**	0.0032**	0.0018**	0.0018**	0.0018**
decile 10	t -statistic	(-3.35)	(-10.84)	(-15.67)	(-22.04)	(-13.90)	(6.97)	(4.10)	(3.60)	(3.62)

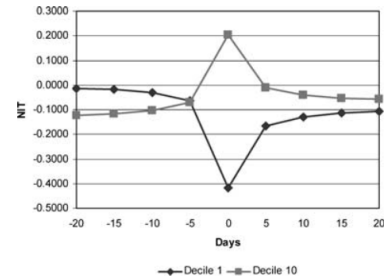


Figure 1. NIT around intense individual trading. This figure presents the Net Individual Trading (NIT) measure of stocks before, during, and after they experience weeks with intense buying and selling activity of individuals. For each week in the sample period, we use the previous 9 weeks to form NIT deciles. Each stock is put into 1 of 10 deciles according to the NIT value in the current week relative to its value in the previous 9 weeks. Decile 1 contains the stocks with the most intense selling (negative NIT) while decile 10 contains the stocks with the most intense buying (positive NIT). Let k be the number of days prior to or following portfolio formation each week. The figures show the average NIT measure of the stocks in decile 1 and 10 during the intense trading week as well as their average NIT in the 4 weeks around the formation week (i.e., $CR(t-k, t-k+4)$, where $k \in \{20, 15, 10, 5\}$ days and t is the first day of the formation week, and $CR(t-k+4, t+k)$, where $k \in \{5, 10, 15, 20\}$ days and t is the last day of the formation week).

5.4 Figure 1: NIT around intense individual trading

Read: The figure plots NIT before, during, and after the intense-trading week for intense buying and intense selling portfolios.

- NIT spikes sharply during the formation week.
- After the formation week, NIT moves back close to its normal level.
- The post-event return drift is therefore not driven by a continued equally intense retail imbalance.

Interpretation. The figure supports the idea that the return pattern is tied to a discrete imbalance event rather than a long persistent wave of retail demand.

5.5 Robustness tests following Table III

Read: The authors report several robustness checks.

They show similar results when they:

- form NIT deciles using cross-sectional sorts rather than stock-specific recent-history sorts;
- use alternative NIT definitions, including unscaled imbalance and buy volume divided by total individual buy-plus-sell volume;
- use market-model, industry-adjusted, raw, and TAQ-midquote returns;
- exclude technology stocks;
- exclude stock-weeks with dividend or earnings announcements.

Interpretation. The NIT return pattern is not an artifact of a single scaling choice, return definition, or sample feature.

5.6 Table IV: Return predictability using portfolio sorting

Read: This table asks whether NIT is just a disguised version of past returns or past turnover.

Panel A: sorting on past return and NIT:

- The NIT spread, buying NIT quintile 5 and selling NIT quintile 1, is positive in every past-return quintile.
- The NIT spread ranges from about 0.24% to 0.60% per week and is statistically significant.
- The past-return spread is not statistically significant after conditioning on NIT.

Panel B: sorting on turnover and NIT:

- The NIT spread is positive and significant in every turnover quintile, roughly 0.29% to 0.42% per week.
- The turnover spread is also positive and significant in every NIT quintile, roughly 0.37% to 0.70% per week.

Interpretation. NIT is not just short-term return reversal. It is also not just high-volume return predictability. NIT and turnover contain different information.

Table IV

Return Predictability: Portfolio Sorting Approach

This table presents analysis of weekly return predictability conditional on the previous week's return (Panel A) or turnover (Panel B) and the net individual trading measure (NIT). For each week in the sample period, we use the previous 9 weeks to form NIT quintiles. Each stock is put into one of the five quintiles according to the NIT value in the current week relative to its value in the previous 9 weeks (where quintile 1 has stocks with more negative NIT, or more selling, and quintile 5 has stocks with more positive NIT, or more buying). In Panel A, each week in the sample period stocks is also sorted on returns and put into five quintiles (quintile 1 has stocks with the most negative returns and quintile 5 has stocks with the most positive returns). We then form 25 portfolios as the intersection of the five return quintiles and five NIT quintiles, and compute for each portfolio the market-adjusted return in the week following the formation week. We present the time-series mean return for each of the 25 portfolios sorted by returns and net individual trading. The last two rows of the panel give the payoff to the strategy of buying NIT quintile 5 and selling NIT quintile 1, and the last two columns of the panel give the payoff to the strategy of buying return quintile 5 and selling return quintile 1. Panel B presents similar analysis except that we place stocks in portfolios based on past turnover (rather than past returns) and past NIT. The construction of the 25 portfolios is analogous to the one in Panel A, and the last two columns of the panel give the payoff to the strategy of buying turnover quintile 5 and selling turnover quintile 1. * indicates significance at the 1% level (against a two-sided alternative). The *t*-statistic is computed using the Newey-West correction.

Panel A: Weekly Return Predictability Using Past Return and NIT

Return(<i>t</i>)						
NIT(<i>t</i>)	Q1 (<0)	Q2	Q3	Q4	Q5 (>0)	Q5-Q1 <i>t</i> -statistic
Q1 (<0)	-0.0022	-0.0019	-0.0007	-0.0007	-0.0016	0.0006 (0.35)
Q2	-0.0010	-0.0012	-0.0011	-0.0013	-0.0026	-0.0016 (-0.90)
Q3	-0.0001	-0.0017	-0.0007	-0.0010	-0.0012	-0.0011 (-0.60)
Q4	0.0024	-0.0002	-0.0001	0.0005	0.0002	-0.0023 (-1.54)
Q5 (>0)	0.0038	0.0008	0.0017	0.0018	0.0022	-0.0016 (-1.00)
Q5-Q1	0.0060**	0.0027**	0.0024**	0.0025**	0.0038**	
<i>t</i> -statistic	(4.87)	(2.92)	(3.38)	(3.26)	(3.43)	

Panel B: Weekly Return Predictability Using Past Turnover and NIT

Turnover(<i>t</i>)						
NIT(<i>t</i>)	Q1 (<0)	Q2	Q3	Q4	Q5 (high)	Q5-Q1 <i>t</i> -statistic
Q1 (<0)	-0.0033	-0.0031	-0.0021	-0.0005	0.0004	0.0037** (2.78)
Q2	-0.0040	-0.0035	-0.0015	-0.0004	0.0015	0.0055** (3.94)
Q3	-0.0041	-0.0021	-0.0012	0.0000	0.0013	0.0054** (3.89)
Q4	-0.0026	-0.0012	0.0012	0.0028	0.0044	0.0070** (5.41)
Q5 (>0)	0.0036	0.0002	0.0008	0.0030	0.0046	0.0040** (2.98)
Q5-Q1	0.0069**	0.0034**	0.0029**	0.0035**	0.0042**	
<i>t</i> -statistic	(2.95)	(3.58)	(3.44)	(3.20)	(3.64)	

Table V

Return Predictability: Fama-MacBeth Approach

This table presents a regression analysis of short-horizon (weekly) return predictability. The dependent variable is weekly return, $\text{Return}(t+1)$, and the independent variables are an intercept, $\text{Return}(t)$, $\text{NITDecile}(t)$, and $\text{TurnoverDecile}(t)$. The TurnoverDecile variable is similar to that in Gervais et al. (2001). It classifies the weekly turnover (number of shares traded over the number of shares outstanding) into 10 deciles by comparing it to the same stock's turnover in the previous 9 weeks. The net individual trading (NIT) measure is described in Section I, and the NITDecile variable is constructed in a similar fashion to TurnoverDecile . We implement a Fama-MacBeth methodology for the regressions: (i) a cross-sectional regression is performed for each week in the sample period and (ii) test statistics are based on the time series of the coefficient estimates. We present the mean coefficient from the weekly regressions, and use the Newey-West correction for the standard errors to compute the *t*-statistics. In Panel A we use CRSP returns, while in Panel B we compute returns using end-of-day quote midpoints from the TAQ database. * indicates significance at the 1% level and * indicates significance at the 5% level (both against a two-sided alternative).

Panel A: CRSP Returns					Panel B: Midquote Returns from the TAQ Database				
Size Groups	Intercept (<i>t</i> -statistic)	Return(<i>t</i>) (<i>t</i> -statistic)	NITDecile(<i>t</i>) (<i>t</i> -statistic)	Turnover Decile(<i>t</i>) (<i>t</i> -statistic)	Size Groups	Intercept (<i>t</i> -statistic)	Return(<i>t</i>) (<i>t</i> -statistic)	NITDecile(<i>t</i>) (<i>t</i> -statistic)	Turnover Decile(<i>t</i>) (<i>t</i> -statistic)
All stocks	0.0035 (1.93) 0.0008 (0.45) -0.0006 (-0.31) -0.0030 (-1.53)	-0.0225** (-3.10)	0.0005** (7.28)	0.0007** (6.87) 0.0007** (7.71)	All stocks	0.0031 (1.73) 0.0006 (0.33) -0.0007 (-0.37) -0.0031 (-1.60) 0.0024 (-2.51)	-0.0132 (-1.91)	0.0005** (7.03)	0.0007** (6.76) 0.0007** (6.54) 0.0007** (7.44)
Small stocks	0.0028 (1.22) -0.0008 (-0.39) -0.0028 (-1.28)	-0.0312** (-3.93)	0.0006** (6.50)	0.0010** (7.97)	Small stocks	0.0011 (0.52) -0.0029 (-1.30) -0.0057* (-2.57) 0.0039* (2.18) 0.0022 (1.21)	-0.0122 (-1.76) -0.0177* (-2.53)	0.0004** (6.84)	0.0006** (6.19)
Mid-cap stocks	-0.0057* (-2.56) 0.0042* (2.34) 0.0028 (1.30)	-0.0316** (-3.98)	0.0005** (5.64)	0.0010** (8.38)	Mid-cap stocks	0.0011 (0.52) -0.0029 (-1.30) -0.0057* (-2.57) 0.0039* (2.18) 0.0022 (1.21)	-0.0180* (-2.54) -0.0016 (-0.16)	0.0005** (5.71)	0.0009** (8.87)
Large stocks	0.0004 (0.19) 0.0035* (1.99) 0.0015 (0.82) 0.0016 (0.77)	-0.0033 (-0.14) -0.0203 (-1.68)	0.0004** (5.36)	0.0003** (3.18) 0.0003** (3.34)	Large stocks	0.0028 (1.29) 0.0003 (0.15) 0.0032 (1.79) 0.0013 (0.66) 0.0013 (0.62) -0.0007 (-0.36)	0.0005 (0.85) -0.0167 (-1.38)	0.0004** (5.06)	0.0003** (2.97) 0.0003** (3.02)
(continued)									

5.7 Table V: Return predictability using Fama–MacBeth regressions

Read: This table runs cross-sectional weekly regressions of next-week returns on current returns, NIT deciles, and turnover deciles.

Panel A: CRSP returns:

- NITDecile is positive and highly significant in univariate and multivariate specifications.
- TurnoverDecile is also positive and highly significant.
- Past return is negative and significant for all stocks with CRSP returns, but the effect is driven mainly by small stocks.
- In the all-stock multivariate regression, the NITDecile coefficient is about 0.0004 and the TurnoverDecile coefficient is about 0.0007.

Panel B: TAQ midpoint returns:

- NITDecile remains positive and highly significant.
- TurnoverDecile remains positive and highly significant.
- Past-return reversal weakens substantially and is not significant for all stocks.
- With midpoint returns, past return remains significant mainly for small stocks.

Interpretation. NIT survives standard return-predictability controls. Past return is more fragile, especially when bid–ask bounce is reduced using midpoint returns.

5.8 Table VI: Historical trends in return reversal

Read: This table estimates weekly return reversal regressions over nonoverlapping four-year CRSP windows from 1964 to 2003.

Key pattern:

- The all-stock past-return coefficient is strongly negative in earlier decades.
- It monotonically declines in magnitude from -0.0909 in 1988–1991 to -0.0376 in 1996–1999 and -0.0229 in 2000–2003.
- In 2000–2003, reversal remains significant mostly for small stocks; mid-cap and large stocks have weak or insignificant reversal coefficients.

Interpretation. The classic weekly return-reversal effect appears weaker in the sample period of this paper, especially for larger stocks. This strengthens the case that NIT is not merely reproducing the older reversal effect.

Table VI
Return Predictability: Historical Trends

This table presents an investigation of historical trends in short-horizon (weekly) return predictability with past returns as the predictive variable. The dependent variable is weekly return (from CRSP), $\text{Return}(t+1)$, and the independent variables are an intercept and $\text{Return}(t)$. We implement a Fama-MacBeth methodology for the regressions: (i) a cross-sectional regression is performed for each week in the sample period and (ii) test statistics are based on the time series of the coefficient estimates. We present the mean coefficient from the weekly regressions, and use the Newey-West correction for the standard errors to compute the t -statistics. Since our main analysis (e.g., Table V) uses 4 years of data (2000 to 2003), we examine historical trends by running the regressions on nonoverlapping 4-year periods going back from 2003 to the beginning of data availability in CRSP. The table presents regression results for all stocks and by size groups. We sort stocks according to market capitalization into 10 deciles, and define deciles 1, 2, 3, and 4 as small stocks, deciles 5, 6, and 7 as mid-cap stocks, and deciles 8, 9, and 10 as large stocks. ** indicates significance at the 1% level and * indicates significance at the 5% level (both against a two-sided alternative).

	All Stocks		Small Stocks		Mid-Cap Stocks		Large Stocks	
	Intercept	Return(t)	Intercept	Return(t)	Intercept	Return(t)	Intercept	Return(t)
1964–1967	0.0039**	-0.0765**	0.0054**	-0.0925**	0.0036**	-0.0695**	0.0024*	-0.0561**
	(3.21)	(-11.33)	(3.77)	(-12.32)	(2.95)	(-8.31)	(2.23)	(-7.27)
1968–1971	0.0013	-0.0929**	0.0013	-0.1084**	0.0013	-0.0848**	0.0012	-0.0786**
	(0.63)	(-12.63)	(0.58)	(-12.83)	(0.64)	(-9.87)	(0.72)	(-10.05)
1972–1975	0.0004	-0.0973**	0.0006	-0.1263**	0.0004	-0.0814**	0.0003	-0.0635**
	(0.16)	(-14.59)	(0.22)	(-17.86)	(0.16)	(-10.24)	(0.13)	(-7.64)
1976–1979	0.0046**	-0.0797**	0.0062**	-0.0930**	0.0046**	-0.0804**	0.0023	-0.0658**
	(3.04)	(-12.58)	(3.33)	(-13.98)	(3.06)	(-10.88)	(1.78)	(-9.06)
1980–1983	0.0051**	-0.0698**	0.0061**	-0.0765**	0.0050**	-0.0715**	0.0042**	-0.0657**
	(3.04)	(-13.34)	(3.38)	(-13.49)	(2.98)	(-10.67)	(2.52)	(-7.85)
1984–1987	0.0023	-0.0688**	0.0013	-0.0758**	0.0026	-0.0720**	0.0035	-0.0710**
	(1.10)	(-10.54)	(0.58)	(-10.56)	(1.26)	(-9.16)	(1.83)	(-7.89)
1988–1991	0.0036*	-0.0909**	0.0033	-0.1114**	0.0033*	-0.0358**	0.0036*	-0.0471**
	(2.16)	(-7.83)	(1.64)	(-7.06)	(2.19)	(-4.37)	(2.51)	(-5.31)
1992–1995	0.0031**	-0.0739**	0.0035**	-0.0936**	0.0026**	-0.0331**	0.0029**	-0.0446**
	(3.37)	(-12.63)	(3.14)	(-11.59)	(2.92)	(-4.50)	(3.57)	(-6.42)
1996–1999	0.0028	-0.0376**	0.0022	-0.0448**	0.0029	-0.0182	0.0033*	-0.0392**
	(1.74)	(-5.69)	(1.27)	(-6.75)	(1.72)	(-1.48)	(2.22)	(-5.52)
2000–2003	0.0031	-0.0229**	0.0038*	-0.0383**	0.0033	0.0099	0.0023	-0.0126
	(1.78)	(-3.27)	(1.98)	(-4.94)	(1.86)	(1.09)	(1.30)	(-0.99)

Table VII
Return Volatility around Individual Trading

This table presents analysis of daily standard deviation of returns around intense buying and selling activity of individuals as given by the net individual trading measure (NIT). For each week in the sample period, we use the previous 9 weeks to form NIT deciles. Each stock is put into 1 of 10 deciles according to the NIT value in the current week relative to its value in the previous 9 weeks. Decile 1 contains the stocks with the most intense selling (negative NIT) while decile 10 contains the stocks with the most intense buying (positive NIT). We present the results for four portfolios: (i) decile 1, (ii) deciles 1 and 2, (iii) deciles 9 and 10, and (iv) decile 10. For each stock and each week, we calculate the standard deviation of daily returns in a 9-day window centered on day $k \in \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$, where $k = 0$ is the middle of the formation week. We subtract from these numbers the "normal" 9-day return standard deviation (which we compute as the average of daily return standard deviations on all nonoverlapping 9-day windows in the sample period). Every week we calculate the average of these standard deviations across all the stocks in each of the four portfolios. Each cell in the table contains the time-series mean for each portfolio and a t -statistic testing the hypothesis of a zero mean. The last three columns provide the difference in standard deviations from $k = -20$ to $k = 0$, $k = 0$ to $k = +20$, and $k = -20$ to $k = +20$, with t -statistic testing the hypothesis of zero difference. ** indicates significance at the 1% level and * indicates significance at the 5% level (both against a two-sided alternative).

Portfolio	$k = -20$	$k = -15$	$k = -10$	$k = -5$	$k = 0$	$k = +5$	$k = +10$	$k = +15$	$k = +20$	$k = -20$ to $k = 0$	$k = 0$ to $k = +20$	$k = -20$ to $k = +20$
Intense selling (decile 1)	-0.0001	-0.0002	-0.0001	0.0012**	0.0018**	0.0002	-0.0006	-0.0007	-0.0008	0.0020**	-0.0027**	-0.0007
Selling deciles 1 and 2	-0.209	(-0.38)	(-0.39)	2.60	(4.22)	(0.45)	(-1.30)	(-1.73)	(-1.91)	(4.41)	(-5.53)	(-1.34)
Buying deciles 9 and 10	0.0001	0.0001	0.0002	0.0008	0.0009*	-0.0002	-0.0006	-0.0007	-0.0008	0.0008	-0.0036**	-0.0009
Intense buying (decile 10)	-0.0008	-0.0008*	-0.0004	0.0005	0.0013**	0.0007	0.0001	0.0000	-0.0001	0.0020**	-0.0014*	0.0006
	(-1.89)	(-2.05)	(-1.09)	(1.22)	(2.66)	(1.57)	(0.18)	(0.04)	(-0.28)	(4.05)	(-2.91)	(1.23)
Intense trading	-0.0010**	-0.0011**	-0.0008*	0.0008	0.0022**	0.0013**	0.0003	0.0001	-0.0001	0.0033**	-0.0023**	0.0010
	(-2.62)	(-2.83)	(-2.00)	(1.83)	(4.58)	(2.68)	(0.62)	(0.17)	(-0.15)	(6.44)	(-4.71)	(1.88)

5.9 Table VII: Return volatility around individual trading

Read: This table asks whether the NIT return premium reflects increased risk.

Key numbers:

- Around intense selling, abnormal volatility rises by about 0.0020 from event time $k = -20$ to $k = 0$, then falls by about 0.0027 from $k = 0$ to $k = +20$.
- Around intense buying, abnormal volatility rises by about 0.0033 from $k = -20$ to $k = 0$, then falls by about 0.0023 from $k = 0$ to $k = +20$.
- The long-run difference from $k = -20$ to $k = +20$ is not statistically significant in the main intense-trading portfolios.

Interpretation. Volatility rises around intense trading but does not stay permanently elevated. This makes a risk-compensation explanation less plausible.

5.10 Liquidity evidence using effective-spread sorts

Read: The authors sort stocks by percentage effective spread and repeat the intense-NIT return analysis.

Most important number:

- For intense individual buying, the 20-day post-event excess return is about 0.42% in low-spread stocks, 0.59% in medium-spread stocks, and 1.40% in high-spread stocks.

Interpretation. The return to taking the other side of order flow is larger when liquidity is scarce. This is exactly what a liquidity-provision story predicts.

5.11 Figure 2: Returns following intense individual trading

Read: The figure plots cumulative market-adjusted returns up to 50 days after intense individual trading.

- The post-buying cumulative return rises over roughly the first 20 days.
- The post-selling cumulative return falls over roughly the first 20 days.
- The effect stabilizes after about 20 days.

Interpretation. The return effect is short horizon. This pattern fits temporary price pressure better than a long-run private-information story.

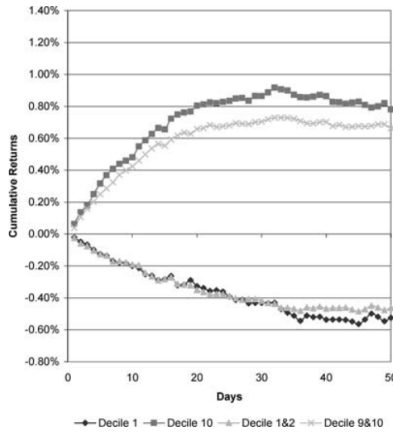


Figure 2. Returns following intense individual trading. This figure presents cumulative market-adjusted returns following weeks with intense buying and selling activity of individuals as given by the net individual trading measure (NIT). For each week in the sample period, we use the previous 9 weeks to form NIT deciles. Each stock is put into 1 of 10 deciles according to the NIT value in the current week relative to its value in the previous 9 weeks. Decile 1 contains the stocks with the most intense selling (negative NIT) while decile 10 contains the stocks with the most intense buying (positive NIT). We present the results for four portfolios: (i) decile 1, (ii) deciles 1 and 2, (iii) deciles 9 and 10, and (iv) decile 10. We calculate cumulative return numbers for each of the stocks in a portfolio: $CR(t+1, t+k)$, where t is the last day of the portfolio formation week and k is the number of days in the cumulative return calculation. The return on each portfolio is then adjusted by subtracting the return on a market proxy (the equal-weighted portfolio of all stocks in the sample).

Table VIII
Summary Statistics of the Aggregate Net Individual Trading Measure

The sample of stocks for the study consists of all common domestic stocks that were traded on the NYSE at any time between January 1, 2000, and December 31, 2003, with records in the CRSP database. We use ticker symbol and CUSIP to match the stocks to a special data set containing daily aggregated buying and selling volume of individuals provided by the NYSE. There are 2,034 stocks in our sample. Stocks are sorted by market capitalization and put into three groups: small (deciles 1, 2, 3, and 4), mid-cap (deciles 5, 6, and 7), and large (deciles 8, 9, and 10). For each size group and for the entire market we compute the dollar imbalance of individuals aggregated across all stocks in the group, and provide time-series summary statistics for this dollar imbalance measure.

	Mean (million \$)	Std. Dev. (million \$)	Min (million \$)	25% (million \$)	Median (million \$)	75% (million \$)	Max (million \$)
Entire sample	-240.58	170.98	-1,063.72	-316.19	-217.38	-132.51	1,690.68
Small stocks	-87.78	7.96	-40.29	-13.85	-7.62	-3.05	20.44
Mid-cap stocks	-28.54	19.36	-115.77	-39.86	-27.48	-14.17	31.70
Large stocks	-203.27	157.28	-956.59	-269.45	-184.80	-107.48	1,740.01

5.12 Table VIII: Aggregate net individual trading

Read: This table aggregates individual buying and selling across all sample stocks each day.

Important numbers:

- The mean aggregate imbalance for the entire sample is about -\$240.58 million per day.
- The median aggregate imbalance is about -\$217.38 million per day.
- The maximum aggregate imbalance is about +\$1.69 billion, showing that aggregate retail imbalances can be large.
- Large stocks account for most of the aggregate dollar imbalance.

Table IX

Principal Components Analysis

This table presents a principal components analysis of returns and the net individual trading measure (NIT) at the daily frequency. Panel A reports the results of a principal components analysis of 1,000 subsamples of 180 stocks each (since we have more stocks in our sample than days in the sample period). We perform a principal components analysis on each subsample, and report the mean (Real Mean) and standard deviation (Real Std.) across subsamples of the percentage of the variance explained by the first 10 principal components. We then construct 1,000 additional 180-stock random subsamples. We compute for each stock the mean and standard deviation of the variable of interest (say, NIT) and generate an artificial time series for each stock drawn from a normal distribution with the same mean and standard deviation. We perform a principal components analysis on the simulated data of each subsample, and report the mean (Simulated Mean) across subsamples of the percentage of the variance explained by the first 10 principal components. We then report the difference in the percentage of the variance explained by the different principal components (PC1, PC2, sum of PC1–5, sum of PC1–10) between the real data and the simulated data. Panel B reports the results of a principal components analysis of NIT done separately on each size decile. We sort the stocks according to average market capitalization over the sample period into 10 deciles. We perform a principal components analysis on each decile and report the percentage of the variance explained by both the first 5 and the first 10 principal components (PC1–5 and PC1–10, respectively). We then use the mean and standard deviation of each stock to generate 500 artificial time series drawn from the normal distribution to form 500 independent subsamples for each decile. We perform a principal components analysis on each subsample and save the mean across the subsamples of the percentage of the variance explained by the first 5 and 10 principal components. We then report the difference in the percentage of the variance explained by the principal components between the real data and the simulated data.

Panel A: Percentage of Variance Explained by Principal Components (1,000 Random Samples of 180 Stocks)

		PC1	PC2	PC1–5	PC1–10
Returns	Real mean	0.1317	0.0297	0.2125	0.2709
	Real std.	0.0079	0.0027	0.0097	0.0100
	Simulated mean	0.0111	0.0108	0.0533	0.1033
	Difference	0.1207	0.0189	0.1592	0.1676
NIT	Real mean	0.0280	0.0239	0.1007	0.1641
	Real std.	0.0022	0.0017	0.0044	0.0052
	Simulated mean	0.0111	0.0108	0.0533	0.1033
	Difference	0.0170	0.0131	0.0474	0.0608

Panel B: Percentage of Variance of NIT Explained by Principal Components (Size Deciles)

		Decile 1 (small)	Decile 2	Decile 3	Decile 4	Decile 5	Decile 6	Decile 7	Decile 8	Decile 9	Decile 10 (large)
PC	Real	0.0805	0.0945	0.0872	0.0941	0.0946	0.0994	0.0995	0.1096	0.1202	0.1521
	Diff	0.0313	0.0454	0.0382	0.0450	0.0456	0.0503	0.0504	0.0606	0.0710	0.1031
PC	Real	0.1427	0.1536	0.1456	0.1552	0.1528	0.1558	0.1573	0.1676	0.1846	0.2191
	Diff	0.0474	0.0583	0.0506	0.0599	0.0576	0.0605	0.0619	0.0727	0.0893	0.1241

Interpretation. Aggregate imbalances can be economically large. But large aggregate imbalance does not by itself prove systematic retail sentiment; it could come from a few stocks with large imbalances.

5.13 Table IX: Principal components analysis

Read: This table asks whether NIT has a strong common component across stocks.

Panel A:

- For daily returns, the first principal component explains 13.17% of variance in the real data, versus 1.11% in simulated independent data. The adjusted difference is 12.07%.
- For NIT, the first principal component explains 2.80% in the real data, versus 1.11% in simulated data. The adjusted difference is only 1.70%.
- The first five components explain 21.25% of return variance in real data, but only 10.07% of NIT variance in real data.

Panel B:

- The common component in NIT is not especially strong among small stocks.
- Adjusted PC1–PC5 explanatory power is higher in large stocks than in small stocks.

Interpretation. The paper does not find strong evidence that individual trading is a market-wide systematic noise factor. NIT matters mainly as a stock-level liquidity-pressure signal.

6 How the paper fits into the literature

6.1 Individual investors as contrarians

The paper is consistent with a large international literature showing that individuals often trade against recent short-run returns:

- buying after prices decline,
- selling after prices rise.

This pattern appears in studies of Korea, Finland, Australia, Asian markets, Nasdaq retail flow, and U.S. index-fund investors.

6.2 Why return effects differ across papers

The paper argues that different findings in the literature can depend on:

- the horizon studied: days/weeks versus months/years;
- the market structure: institution-dominated markets versus individual-dominated markets;
- the data source: direct account-type data, brokerage records, or signed small trades;
- whether the researcher studies all retail trades or only intense imbalances.

Key reconciliation. In institution-dominated markets such as the United States and Australia, contrarian individuals may profit at short horizons by providing liquidity. In markets dominated by individuals, such as Taiwan, that liquidity-provision role may be less relevant.

6.3 Relation to small-trade proxies

Some papers infer individual trading from small trades. The advantage is a long time series. The disadvantage is measurement error, especially as institutions increasingly split large orders into small trades.

The NYSE account-type data are cleaner for identifying individuals, but the sample period is shorter.

7 Short summary

- **Object.** Net Individual Trading (NIT) equals individual buy dollar volume minus individual sell dollar volume, scaled by past average dollar volume.
- **Data.** The paper uses proprietary NYSE order data from 2000–2003 for 2,034 common domestic NYSE stocks, matched to CRSP and TAQ.
- **Behavioral pattern.** Individuals are contrarian at short horizons: they buy after recent price declines and sell after recent price increases.
- **Return predictability.** Intense individual buying predicts positive next-month abnormal returns; intense individual selling predicts negative next-month abnormal returns.
- **Controls.** NIT remains predictive after controlling for past returns and turnover through both portfolio sorts and Fama–MacBeth regressions.
- **Risk test.** Volatility rises around intense trading but does not remain permanently high, so the return premium is unlikely to be pure risk compensation.
- **Liquidity test.** The effect is stronger in less liquid stocks, supporting the interpretation that individuals provide liquidity to institutions demanding immediacy.
- **Systematic noise.** Individual trading does not have a strong common component across stocks; NIT is mostly a stock-level signal rather than a broad market sentiment factor.

8 Conclusion

8.1 Core conclusion

Kaniel, Saar, and Titman show that aggregate individual-investor trading on the NYSE predicts short-horizon stock returns. Stocks heavily bought by individuals earn positive abnormal returns afterward, while stocks heavily sold by individuals earn negative abnormal returns afterward.

8.2 Economic intuition

The main mechanism is liquidity provision. Institutions sometimes need immediacy and push prices away from fundamentals. Individuals, by trading contrarian, absorb the other side of this order flow. The price concession they receive shows up as a short-run reversal.

8.3 Takeaway

The paper changes how one should think about individual investors. They may not be sophisticated information traders, and they may not always perform well over long horizons. But their aggregate contrarian trades can still be economically useful: they supply liquidity and reveal where temporary price pressure is likely to reverse.